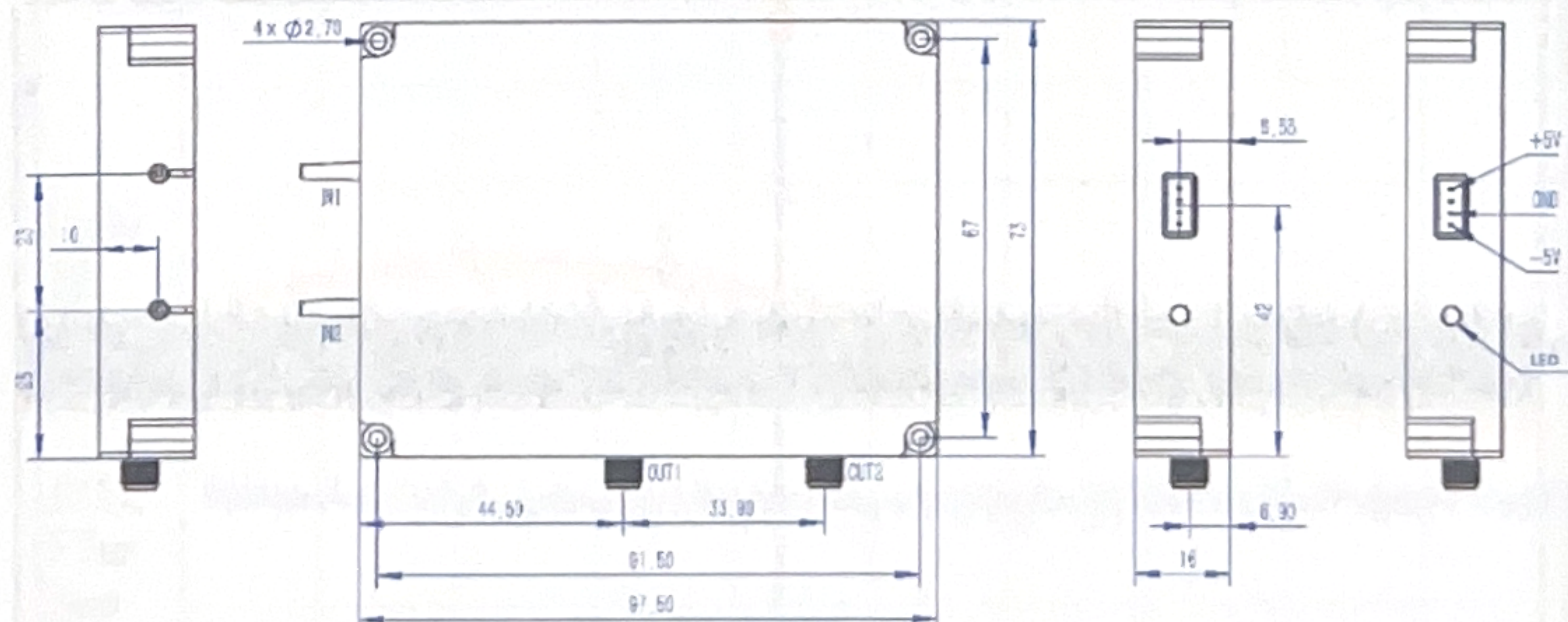


APD-100-35-FA Avalanche Photodetector Test Report

Serial Number: R001000010

Parameters	Value
Detector Type	InGaAs APD
Wavelength Range	900 – 1700 nm
Output Bandwidth (3dB)	DC – 33MHz
Active Area Diameter	50 μm
Responsivity (Square Waveform)	120mVp-p/0.01uW
Sensitivity (Noise Ratio 1:1)	-63dBm
NEP	0.087pW/ $\sqrt{\text{Hz}}$
Dynamic Range (Linear Gain)	27dB
Electrical Noise	10mV _{p-p}
CW Saturation Power	-35dBm
Max Output Voltage Swing (Square Waveform)	+2.5V _{p-p}
Power Supply	Red Wire: +5/1A, White Wire: -5V/0.1A, Shielded Wire: Ground
Fiber Type	SM Fiber, FC/APC
Operating Temperature	0 - 40°C
Storage Temperature	-40 - 70°C
Dimensions	97.5 (L) x 73 (W) x 16 (H) mm

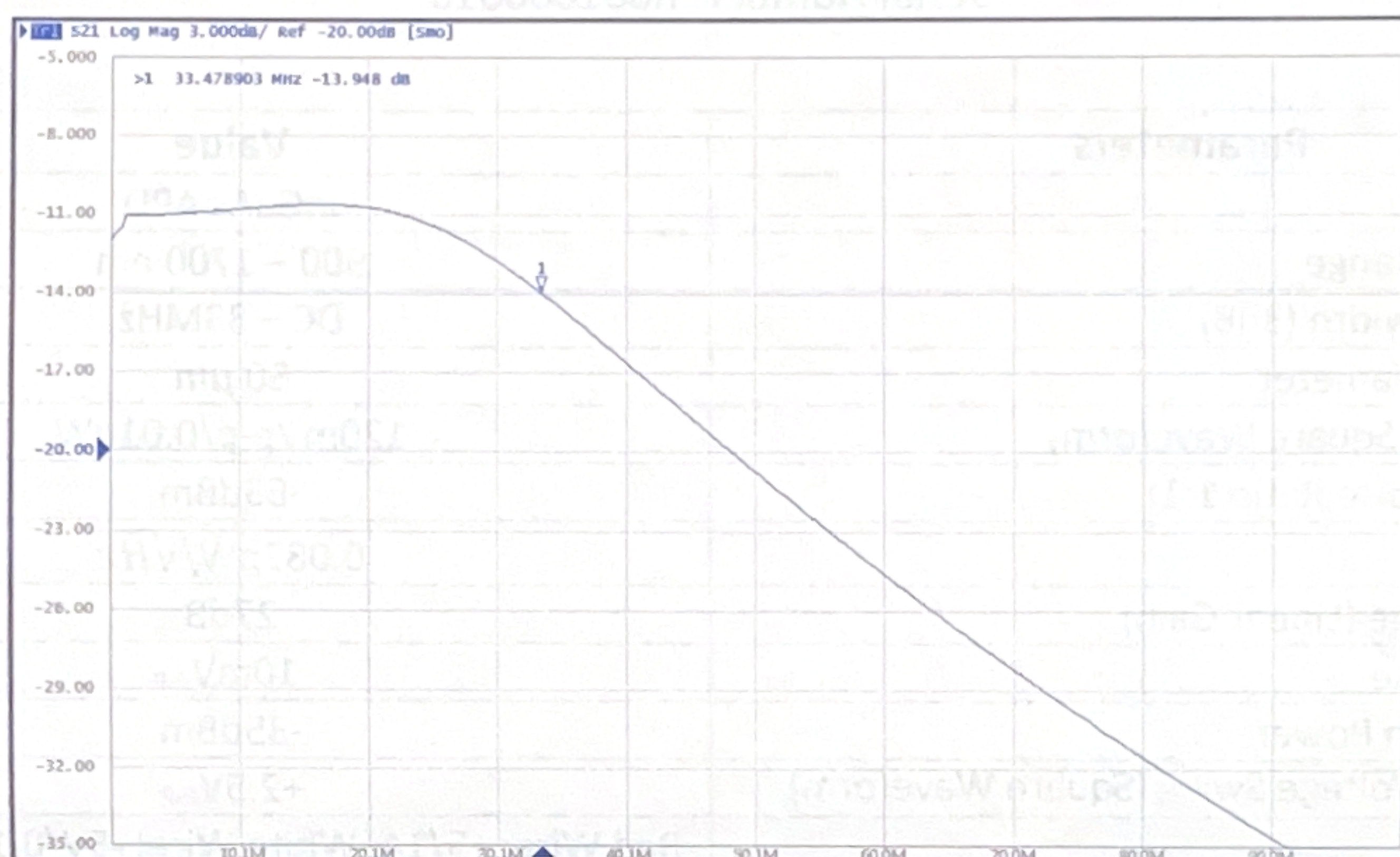


Test By: Hogan jia

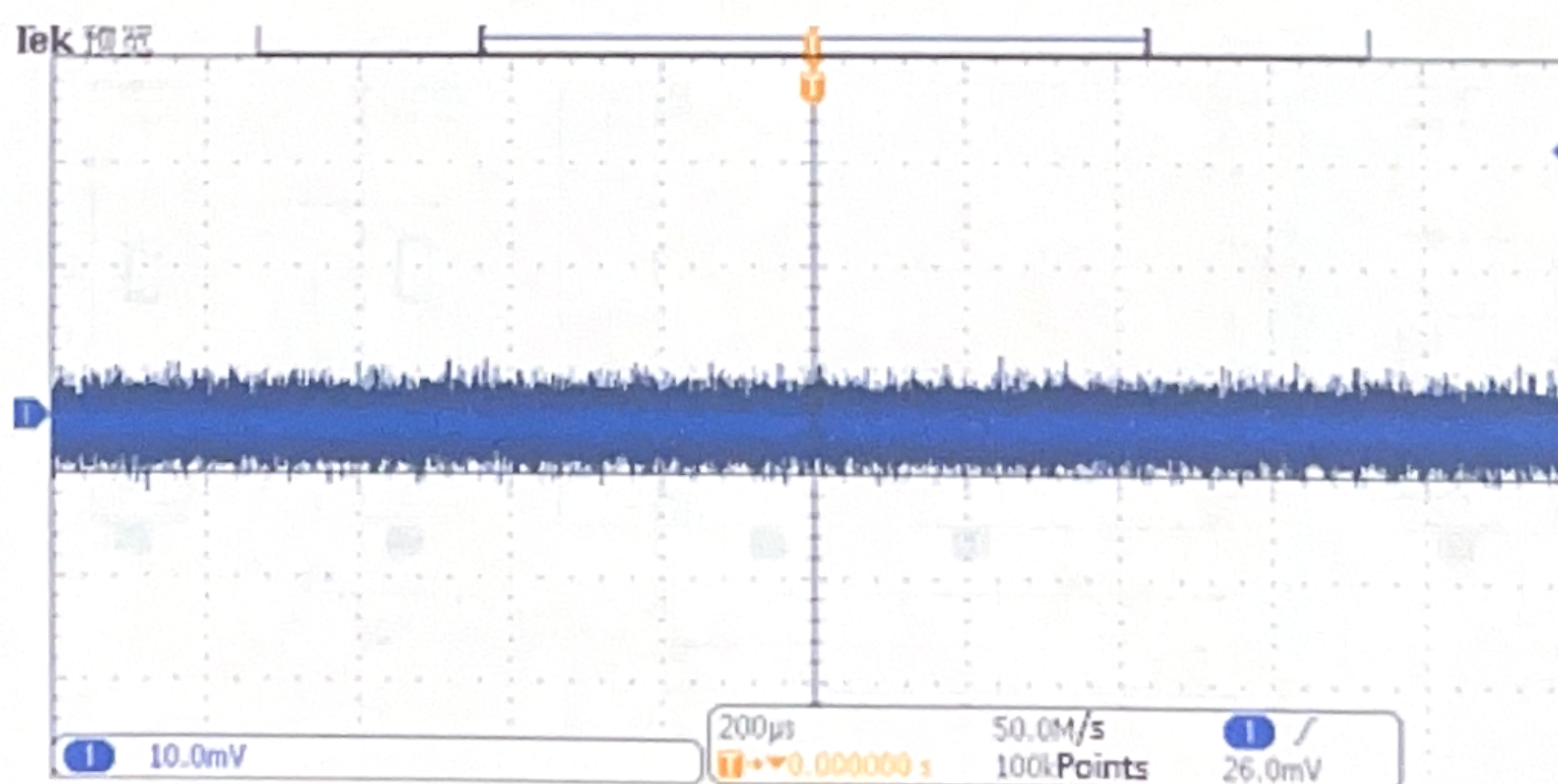
Date: 2024/11/14

Appendix:

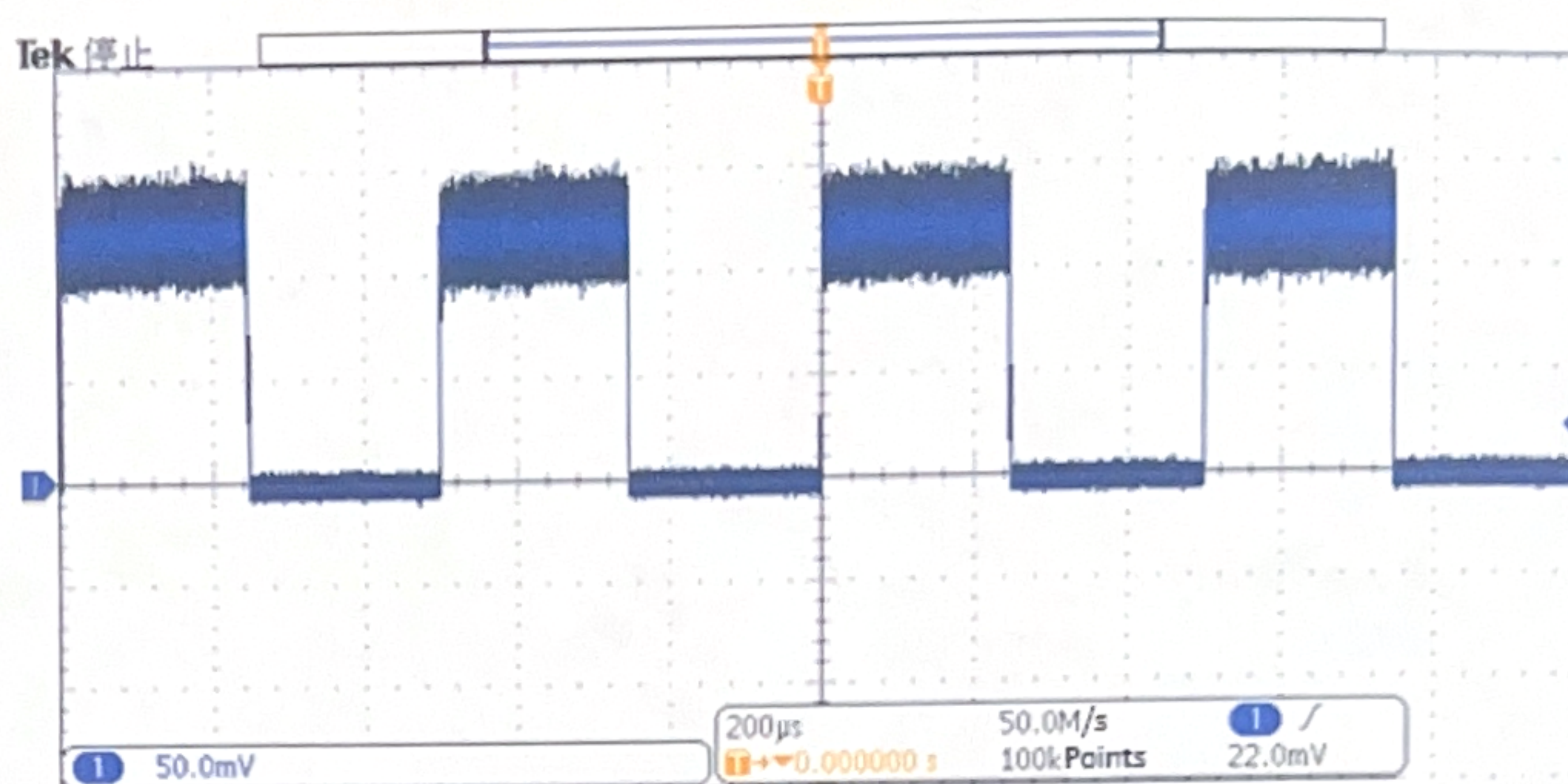
1. Amplitude-Frequency Characteristics (33MHz):



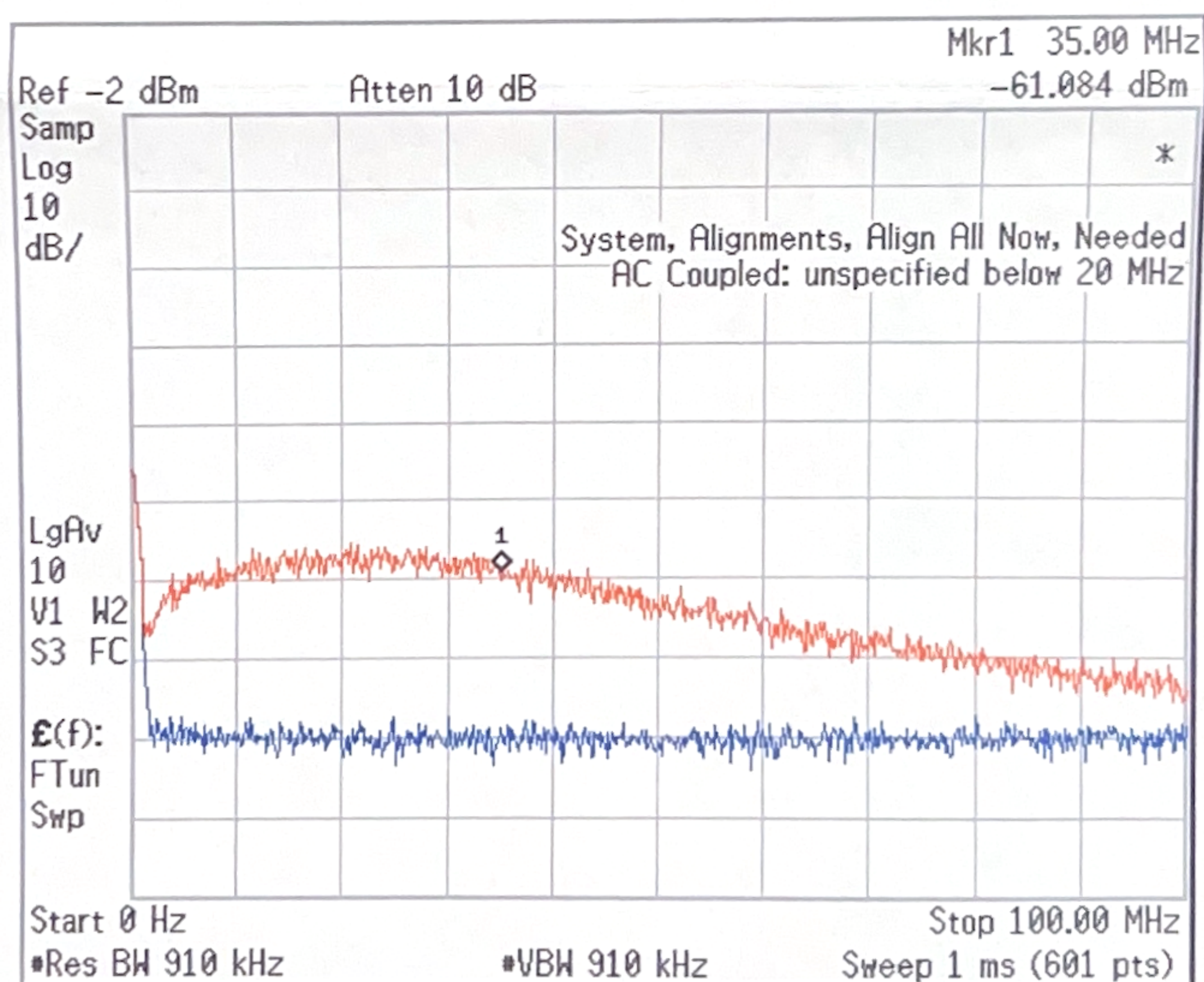
2. Electrical Noise (10mVp-p: Oscilloscope impedance 1M Ω):



3. Responsivity (Square Waveform) (120mVp-p/-50.0dBm: Oscilloscope impedance 1M Ω):



4. Frequency Spectrum (Blue line is Frequency Spectrum Analyzer background noise):



APD-100 Application Notes

1. Input Light

The damage threshold for APD (Avalanche Photodiode) input optical power is required to be below **10 μ W**. Exceeding this value will permanently damage the detector! When connecting the optical signal, ensure the cleanliness of the fiber connector end face and that the fiber connector is tightly secured. Failure to do so will reduce the detector's responsiveness and decrease the amplitude of the output signal.

2. Power Supply

For applications requiring low noise, use a linear power supply. If a switching power supply is used, the ripple from the power supply may couple into the detector's input stage, causing significant ripple interference. The module requires ± 5 V DC power supply:

- **The +5V voltage range is +5.0 to +5.5V/1A.** Since the +5V is used for the internal temperature control circuit, when the detector is powered on, a considerable temperature control current is required, resulting in higher power consumption. Once it reaches a stable working state at room temperature, the power consumption is less than 0.6W.
- **The -5V voltage range is -5 to -5.5V/0.1mA.** When the power supply is functioning properly, the power indicator LED will light up. Ensure that the power supply polarity is correct and within the specified voltage range to avoid damage.

3. Output

The detector has a high transimpedance gain and sensitivity. If the input light exceeds the linear gain range, the output signal will saturate. Refer to the detector test report for the saturated optical power specification. The module has an output impedance of 50Ω and a maximum output signal amplitude of 2.5V peak-to-peak.

4. APD Test Conditions

Testing conditions: ± 5 V DC power supply, 500K square wave, optical power -36.0 dBm.

Order Information

Invoice No.	241003
Invoice Date	10/15/2024
Customer PO No. (Date)	777577

10/15/2024